

RETAIL SALES & INVENTORY DASHBOARD

Project Report

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Project Summary

The Retail Sales & Inventory Dashboard project is developed using SQL Server and Power BI to analyze retail business data through interactive dashboards and business intelligence techniques. The dashboard provides insights related to total sales, customer analysis, store performance, category-wise sales, and staff performance.

The project demonstrates practical implementation of:

- SQL queries and data management
- Data visualization using Power BI
- KPI analysis
- Interactive filtering using slicers
- Business intelligence reporting

The dashboard helps organizations monitor sales trends, compare store performance, and make better business decisions using data-driven insights.

Skills Used

- SQL Server
- Power BI
- Data Analysis
- Dashboard Design
- Data Visualization
- KPI Reporting
- Business Intelligence
- Database Management
- Interactive Reporting

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Chapter 1: Introduction

The Retail Sales & Inventory Dashboard project is developed to analyze retail business performance using Business Intelligence tools. The dashboard provides visual insights related to sales, customers, staff performance, inventory, and store-wise analysis.

The project uses SQL Server as the database management system and Power BI as the visualization and reporting tool. Data is collected from multiple retail-related tables such as orders, customers, products, stores, categories, and staff.

The main purpose of the dashboard is to transform raw retail data into meaningful visual insights for better business decision-making.

The dashboard enables users to:

- Monitor total sales and orders
- Analyze category-wise sales
- Compare staff performance
- Track store-wise sales
- Analyze sales trends over time
- Apply filters dynamically using slicers

Business Intelligence dashboards help organizations identify trends, improve operational efficiency, and make data-driven decisions.

Chapter 2: Objectives of the Project

The major objectives of this project are:

1. To analyze retail sales data efficiently.
2. To create an interactive dashboard using Power BI.
3. To visualize sales performance using charts and KPI cards.
4. To perform category-wise and store-wise sales analysis.
5. To understand customer and staff performance.
6. To improve business decision-making through visual insights.
7. To implement data filtering using slicers.
8. To generate business reports in an interactive format.

Chapter 3: Scope of the Project

The scope of this project includes:

- Retail business analysis
- Sales monitoring
- Customer analysis
- Store performance analysis
- Staff order analysis
- Inventory monitoring
- Trend analysis using line charts

The dashboard can be used by:

- Store Managers
- Sales Teams
- Business Analysts
- Retail Companies
- Management Teams

The project provides real-time visual understanding of business operations and can be extended further using advanced analytics.

Chapter 4: Tools and Technologies Used

1. SQL Server

SQL Server is used for storing and managing retail data. It is used to perform:

- Table creation
- Data storage
- SQL queries
- Joins
- Aggregation
- Filtering

2. Power BI

Power BI is used to create:

- KPI cards
- Bar charts
- Line charts
- Matrix tables

- Interactive slicers
- Dashboard reports

3. Excel

Excel is used for importing or managing dataset files before loading them into SQL Server.

4. DAX (Data Analysis Expressions)

DAX functions are used in Power BI for creating measures and calculations.

Chapter 5: Database Description

The project database contains multiple retail-related tables.

Main Tables Used

1. Orders Table

Contains customer order information.

Fields:

- order_id
- customer_id
- order_date
- store_id
- staff_id

2. Customers Table

Contains customer details.

Fields:

- customer_id
- first_name
- last_name
- city

3. Stores Table

Contains store details.

- Fields:
- store_id

- store_name
- city
- state

4. Staff Table

Contains staff information.

Fields:

- staff_id
- first_name

5. Categories Table

Contains product categories.

Fields:

- category_id
- category_name

6. Order Items Table

Contains sales information.

Fields:

- order_id
- product_id
- quantity
- list_price

Chapter 6: SQL Operations and Data Processing

SQL queries are used to process and analyze retail data.

Operations Performed

1. SELECT Query

Used to retrieve data.

Example:

```
SELECT * FROM orders;
```

2. JOIN Operations

Used to combine multiple tables.

Example:

```
SELECT o.order_id, c.first_name  
FROM orders o  
JOIN customers c  
ON o.customer_id = c.customer_id;
```

3. GROUP BY

Used for aggregation.

Example:

```
SELECT store_id, COUNT(order_id)  
FROM orders  
GROUP BY store_id;
```

4. SUM Function

Used to calculate total sales.

Example:

```
SELECT SUM(list_price)  
FROM order_items;
```

5. COUNT Function

Used to calculate total orders and customers.

Chapter 7: Dashboard Development in Power BI

The dashboard is created in Power BI using various visuals.

KPI Cards Used

1. Total Sales

Displays total sales amount.

2. Total Orders

Displays total number of orders.

3. Total Customers

Displays unique customer count.

4. Total Stores

Displays total number of stores.

Charts Used

1. Sales by Category

Bar chart used to compare category-wise sales.

2. Sales by Staff

Bar chart used to analyze staff performance.

3. Sales by Store

Bar chart used to compare store-wise sales.

4. Sales Over Time

Line chart used to analyze sales trends.

5. Orders by Staff

Matrix table used to display staff order counts.

Slicers Used

Interactive slicers are added for:

- Category
- Store
- City
- Date

These slicers help users dynamically filter dashboard data.

Chapter 8: KPI and Visual Analysis

Total Sales

The dashboard shows overall business revenue generated from retail sales.

Total Orders

This KPI indicates the total number of orders placed by customers.

Total Customers

Shows the total number of unique customers.

Total Stores

Displays the number of stores included in the analysis.

Category Analysis

The category-wise chart helps identify the best-selling product category.

Staff Analysis

The staff analysis chart compares employee performance based on orders and sales.

Store Analysis

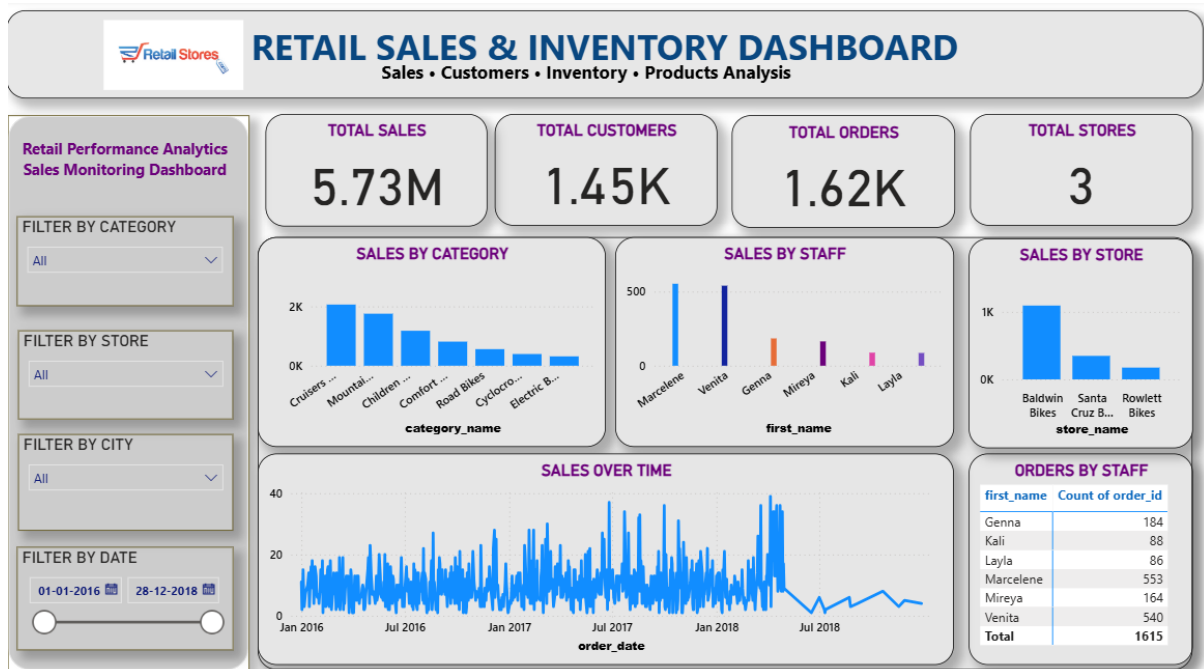
Store-wise comparison helps management identify top-performing stores.

Sales Trend Analysis

The line chart shows fluctuations and trends in sales over time.

Dashboard Screenshots

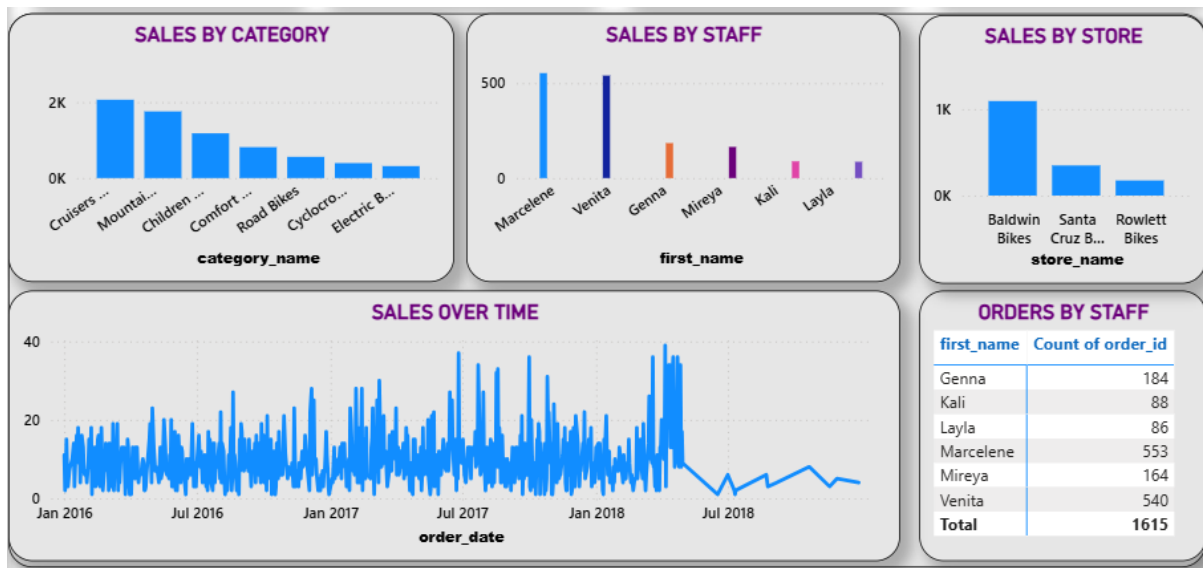
Main Dashboard



KPI Cards



Sales Analysis Charts



Slicers and Filters

**Retail Performance Analytics
Sales Monitoring Dashboard**

FILTER BY CATEGORY

All

FILTER BY STORE

All

FILTER BY CITY

All

FILTER BY DATE

01-01-2016 28-12-2018

Chapter 9: Dashboard Insights

The following business insights were identified from the dashboard:

1. Some product categories generate higher sales than others.
2. Certain staff members handle more customer orders.

3. One store performs significantly better in sales.
4. Sales trends fluctuate across different periods.
5. The dashboard helps management identify performance gaps.

These insights support data-driven business decisions.

Chapter 10: Advantages of the System

1. Interactive dashboard visualization
2. Easy data analysis
3. Real-time filtering using slicers
4. Better business decision-making
5. Improved sales monitoring
6. Staff performance analysis
7. Efficient store comparison
8. User-friendly interface
9. Fast reporting and analysis
10. Professional business presentation

Chapter 11: Limitations

1. Dashboard depends on available dataset quality.
2. Real-time database connection is not implemented.
3. Predictive analysis is not included.
4. Advanced machine learning features are not used.

Chapter 12: Future Enhancements

The project can be improved further by adding:

1. Real-time cloud database integration
2. Machine learning-based sales prediction
3. Inventory forecasting
4. Mobile dashboard support
5. Regional sales maps
6. Advanced drill-through reports
7. Customer segmentation analysis
8. AI-powered business insights

Internship Learning Outcomes

During this internship project, the following practical skills were learned and implemented:

1. Data visualization using Power BI
2. SQL query execution and data analysis
3. Dashboard creation and KPI reporting
4. Business intelligence concepts
5. Interactive filtering using slicers
6. Retail business performance analysis
7. Professional dashboard design
8. Data-driven decision-making concepts

This internship project improved analytical thinking, technical skills, and understanding of real-world business reporting systems.

Chapter 13: Conclusion

The Retail Sales & Inventory Dashboard project successfully demonstrates how Business Intelligence tools can transform raw retail data into meaningful visual insights. The project uses SQL Server for data handling and Power BI for interactive dashboard creation. Various charts, KPI cards, and slicers are used to analyze business performance.

The dashboard helps organizations monitor sales, compare stores, analyze customer behavior, and evaluate staff performance effectively.

This project enhanced practical understanding of:

- SQL queries
- Data analysis
- Power BI dashboard creation
- Data visualization
- Business Intelligence concepts

Overall, the project provides a professional and interactive solution for retail sales analysis.

Chapter 14: References

1. Microsoft Power BI Documentation
2. SQL Server Documentation
3. Microsoft Learn Platform
4. Power BI Tutorials and Training Resources
5. Retail Sales Dataset Documentation
6. Database Management System Concepts

Appendix

Dashboard Components

- KPI Cards
- Bar Charts
- Line Chart
- Matrix Table
- Slicers

Dataset Tables

1. Orders
2. Customers
3. Stores
4. Staff
5. Categories
6. Products
7. Order Items

End of Report